Final System Project Paper

SIA2

HINUNANGAN TOURIST EXPERIENCE SURVEY SYSTEM

Prepared By:
Rose Ann J Sumacot

Course/Section:
BSIT-3B

Instructor:

Jeremiah Dela Cruz

Date Submitted:
May 22, 2026

Introduction

The **Hinunangan Tourism Survey System** is a web-based system designed to help the Municipality of Hinunangan collect, manage, and analyze tourism-related survey data efficiently. Many tourism offices still use manual paper-based surveys, which can result in lost records, inaccurate data, delayed reporting, and difficulty in organizing tourist feedback. Because of these problems, the researchers developed a system that can automate and organize tourism survey management processes.

Background of the Study

Tourism offices and local government units continue to experience challenges in collecting and managing tourism survey data using manual processes. Traditional paper-based surveys are time-consuming, prone to data loss, and difficult to organize, especially when handling large numbers of tourists and visitors. In many municipalities and tourism offices, survey collection and report generation are still performed manually, causing delays in data analysis and inefficient tourism service evaluation.

To address these issues, the **Hinunangan Tourism Survey System** was developed. The system combines tourist information management and survey data collection into one platform to improve efficiency, accessibility, and organization of tourism-related information.

Purpose of the Study/Project

The main purpose of the system is to provide an easier and faster way to manage tourism surveys and tourist information. It digitalizes tourism survey forms and automates the collection and storage of survey responses. The system helps the tourism office monitor tourist feedback properly and generate organized survey reports.

Importance of the Project

This project is important because it improves the efficiency of tourism services by reducing paperwork and minimizing human error in survey collection. It also helps tourism personnel manage tourist feedback more efficiently and accurately while providing faster and more organized services to tourists and visitors. Through this system, the tourism office can better understand tourist experiences and improve tourism management and promotional activities in Hinunangan.

Problems the System Aims to Solve

The **Hinunangan Tourism Survey System** aims to address several problems commonly experienced in tourism offices that still rely on manual survey collection processes. One major problem is the difficulty in managing and organizing tourism survey records, which can result in misplaced files, incomplete responses, and slow retrieval of survey data. Manual record-keeping also increases the risk of data redundancy and human error.

Another problem is the inefficient process of collecting and analyzing tourist feedback. Traditional survey methods may lead to delayed reports, inaccurate records, and difficulty in monitoring tourist satisfaction and tourism activities. These issues can affect the quality and efficiency of tourism services and decision-making.

The system also addresses the problem of excessive paperwork and time-consuming transactions. By digitalizing tourism surveys and tourist information management, tourism personnel can access and manage information more quickly and accurately. Overall, the system aims to improve organization, efficiency, and reliability in handling tourism survey data within the Municipality of Hinunangan.

Objectives of the System

General Objective

To develop an integrated system that manages tourism surveys and tourist information efficiently for the Municipality of Hinunangan.

Specific Objectives

1. To create a digital database for storing tourism survey records.
2. To automate tourism survey collection and management.
3. To reduce paperwork and manual processing.
4. To improve accessibility and retrieval of tourism survey information.
5. To provide secure user authentication for authorized personnel.
6. To improve tourism service efficiency and tourist feedback management.

Scope and Limitations

Scope

The system is designed to:

• Manage tourism survey records
• Store and organize tourist information
• Collect and manage tourism survey responses
• Monitor survey status and reports

• Allow secure login authentication for admin, staff, and tourists
• Enable admin and staff to create tourist accounts
• Allow tourists to answer surveys online without visiting the tourism office
• Update and maintain tourism information efficiently
• Maintain organized and centralized tourism survey records
• Reduce paperwork and improve tourism service efficiency

Limitations

The system is limited to:

• Tourism survey management only
• Authorized users with valid accounts
• Small to medium tourism offices
• Internet-based access for online surveys

The system does not include:

• Online payment transactions
• Hotel or accommodation booking features
• Transportation management features
• SMS or email notification services
• Mobile application support
• Advanced tourism analytics and forecasting functions

Technologies Used

Programming Languages

• HTML – used for creating the structure of web pages
• CSS – used for styling and designing the user interface
• JavaScript – used for adding interactivity and dynamic functions
• PHP – used for backend development and server-side processing

Frameworks/Libraries

• Laravel – used as the main PHP framework for developing the system
• Bootstrap – used for responsive and user-friendly interface design
• JWT Authentication – used for secure user authentication and access control
• Session Management Libraries – used for managing user login sessions

Database Management System

• MySQL Database – used for storing and managing tourism survey and tourist data
• phpMyAdmin – used for database administration and management

Development Tools Used

• Visual Studio Code – used as the primary code editor during development
• GitHub – used for version control and project repository management
• XAMPP – used as the local server environment for running PHP and MySQL services locally during system development

System Design

Use Case Diagram

The Use Case Diagram illustrates the interaction between the users and the system. The main users of the system are the Administrator, Staff, and Tourists.

Administrator

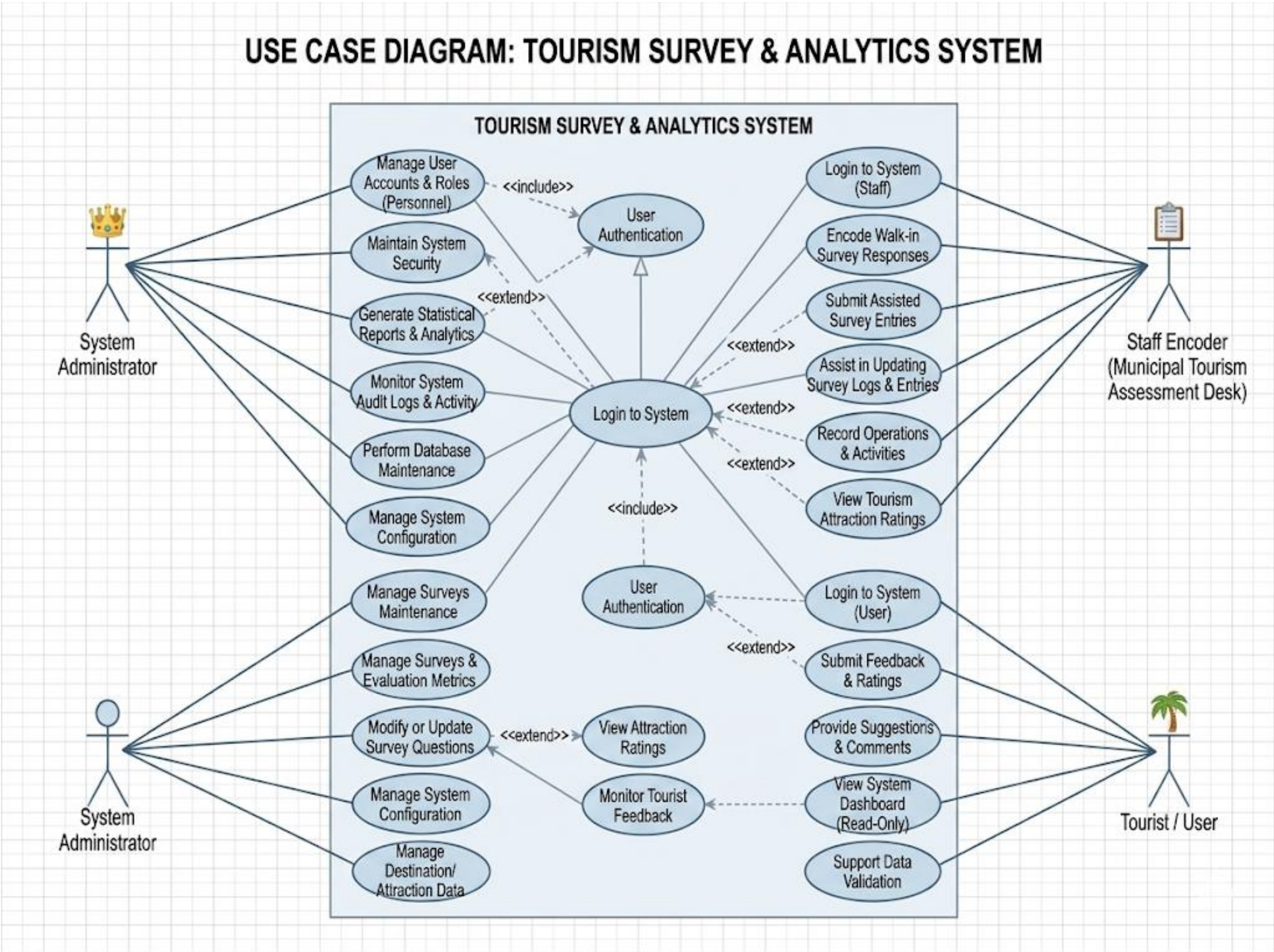
• Manage user accounts
• Manage tourism survey records
• View survey responses
• Update system information

Staff/Tourism Personnel

• Add and update tourist information
• Manage tourism surveys
• Monitor survey responses
• Provide tourist accounts

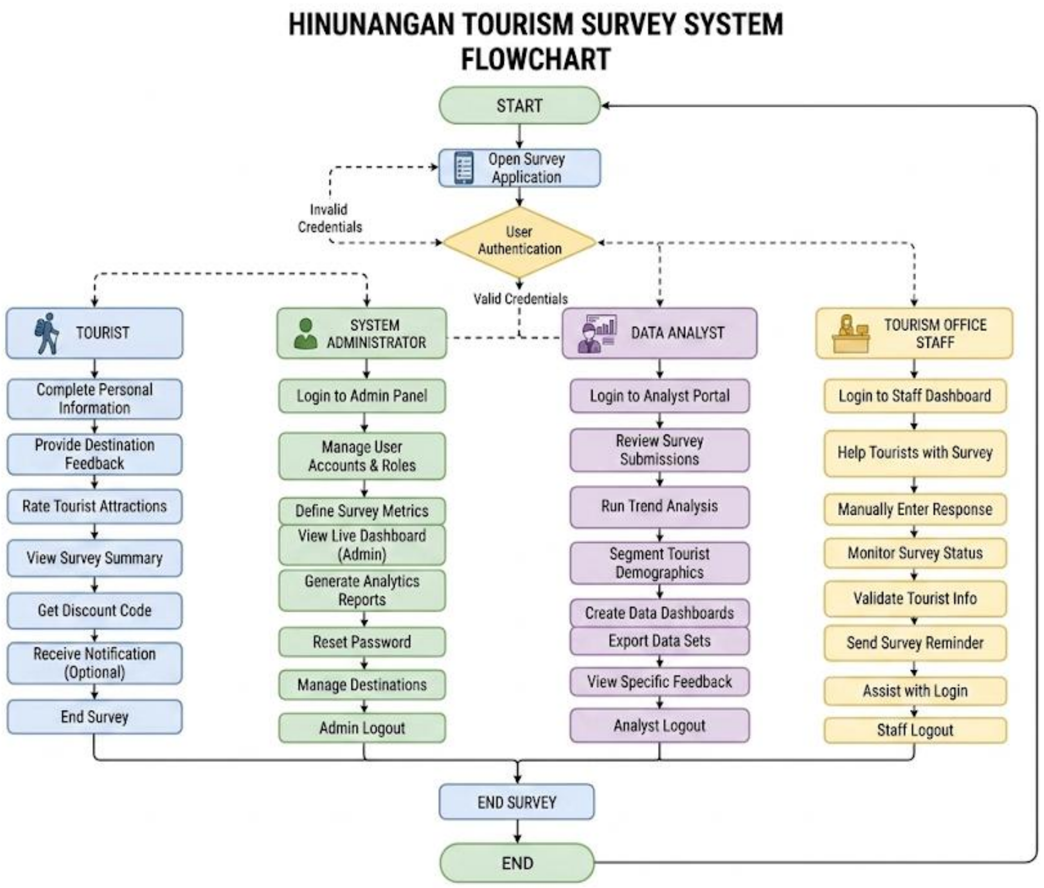
Tourist

• Log in to the system
• View personal information
• Answer tourism surveys online
• View survey status and responses



Flowchart

The system begins when a user accesses the survey page. The user fills out the survey form and submits responses. The system validates the input and stores the data in the database. The administrator can then access the dashboard to view and analyze the collected responses.



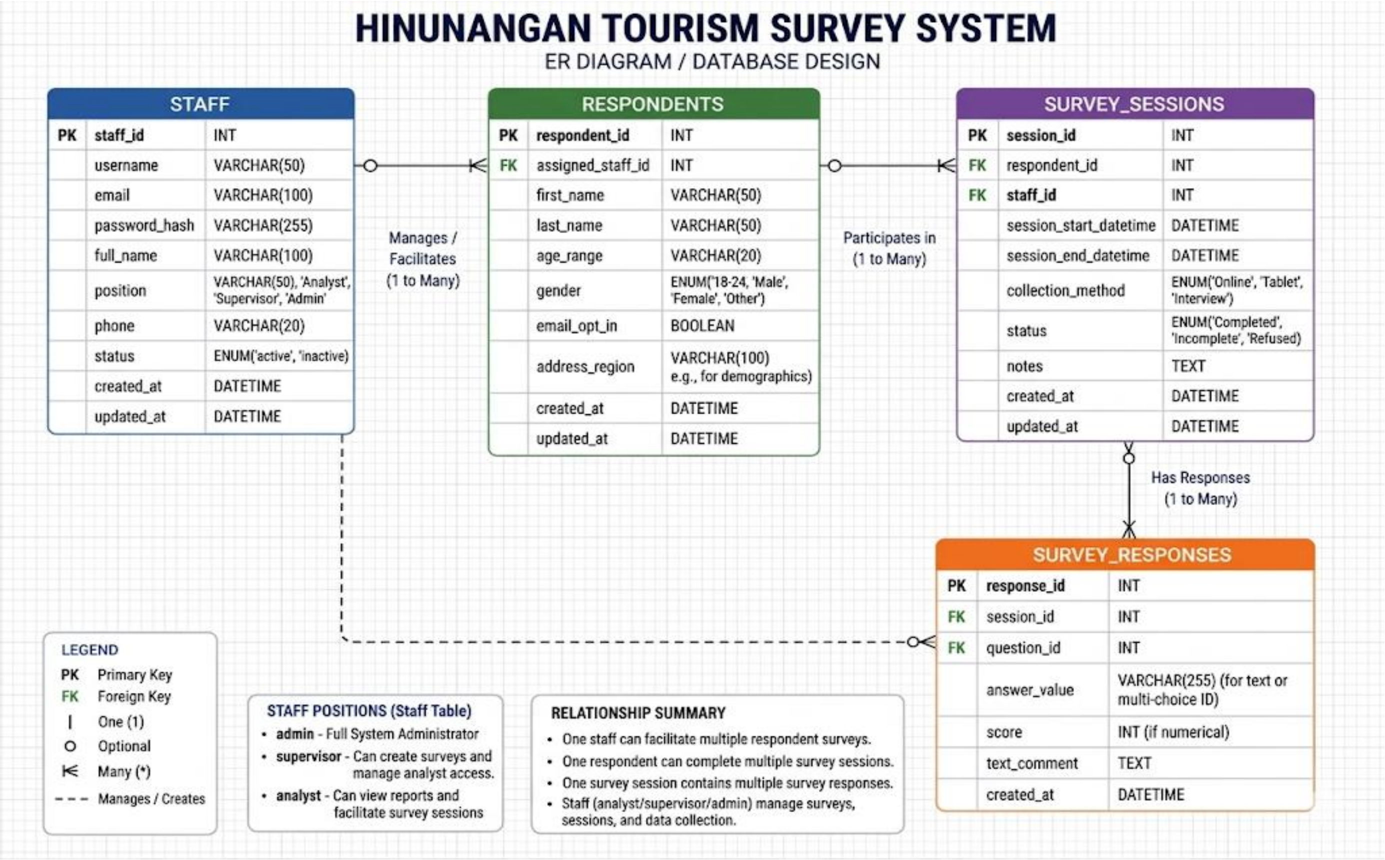
ER Diagram / Database Design

Main Tables:

- Users Table
- Surveys Table
- Questions Table
- Responses Table

Relationships:

- One survey has many questions
- One user can submit many responses
- One survey has many responses



System Architecture Diagram

Client–Server Architecture

1. Client Side (Browser)

- Tourist users
- Admin users

Functions:

- Answer surveys
- View dashboard
- Submit data

2. Application Server (Laravel)

- Handles system logic
- Processes survey submissions
- Manages authentication
- Generates reports

3. Database Server (MySQL)

- Stores survey data
- Stores questions and responses
- Maintains user and admin records

Interface Design / Screenshots

(Insert system screenshots here such as:)

- Survey form page
- Admin dashboard
- Reports page
- Login page

Application Server

Framework: Laravel (PHP Framework)

Main Responsibilities:

- Handles system logic
- Processes user requests from Admin, Staff Encoder, and Tourist/User
- Authentication and authorization of users
- Survey submission processing and validation
- Data analysis and reporting logic
- API routing and request handling
- Implementation of business rules for tourism evaluation system

Components:

- Routes
- Controllers
- Models (Eloquent ORM)
- Middleware
- Blade Templates (UI Rendering)

  The application server acts as the central processing unit of the system, connecting the user interface with the database layer.

Database Server

Database: MySQL

Stores:

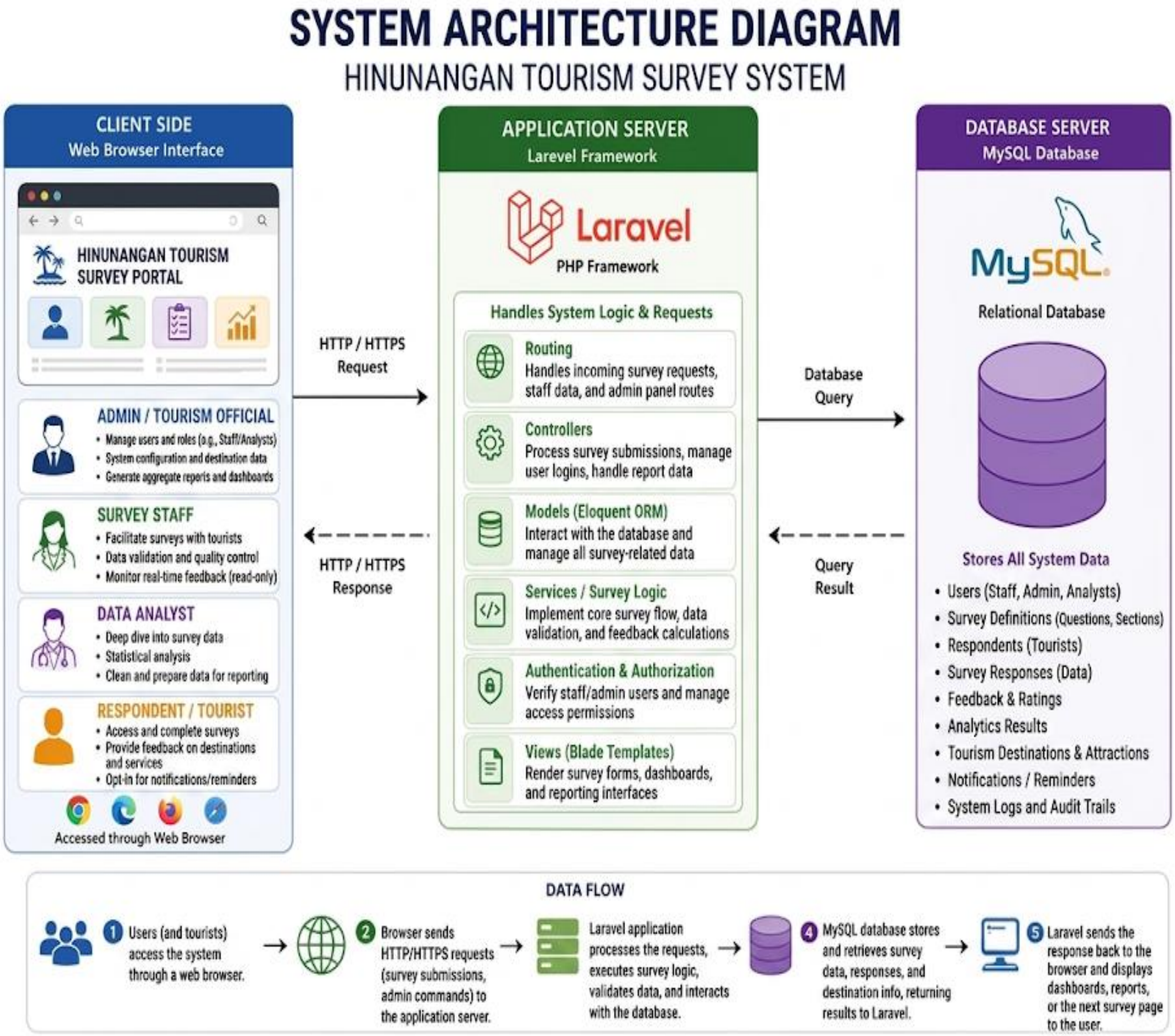
- User accounts (Admin, Staff, Tourist)

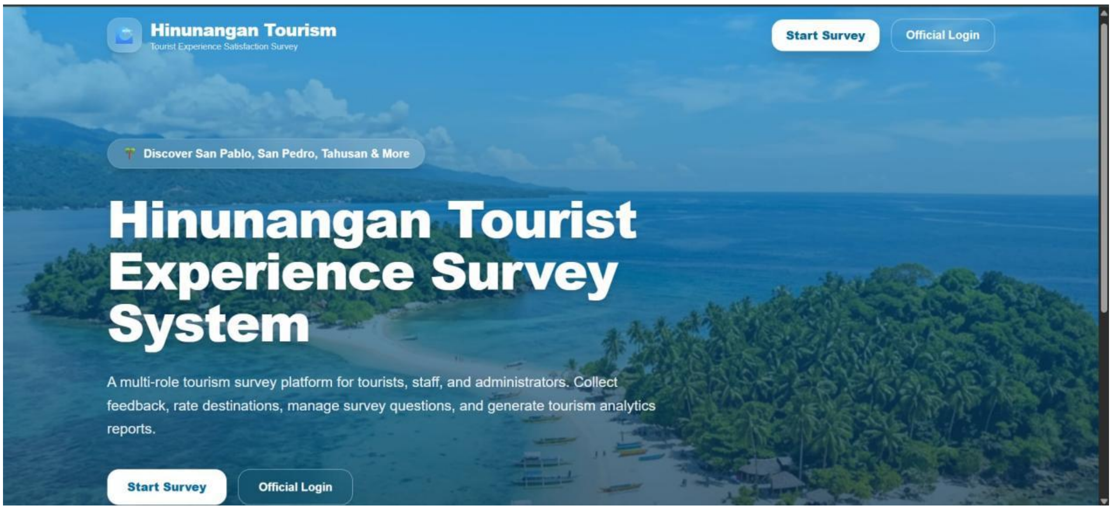
- Tourism survey questions
- Survey responses and feedback
- Tourist destination data
- Satisfaction ratings per location
- System logs and activity records

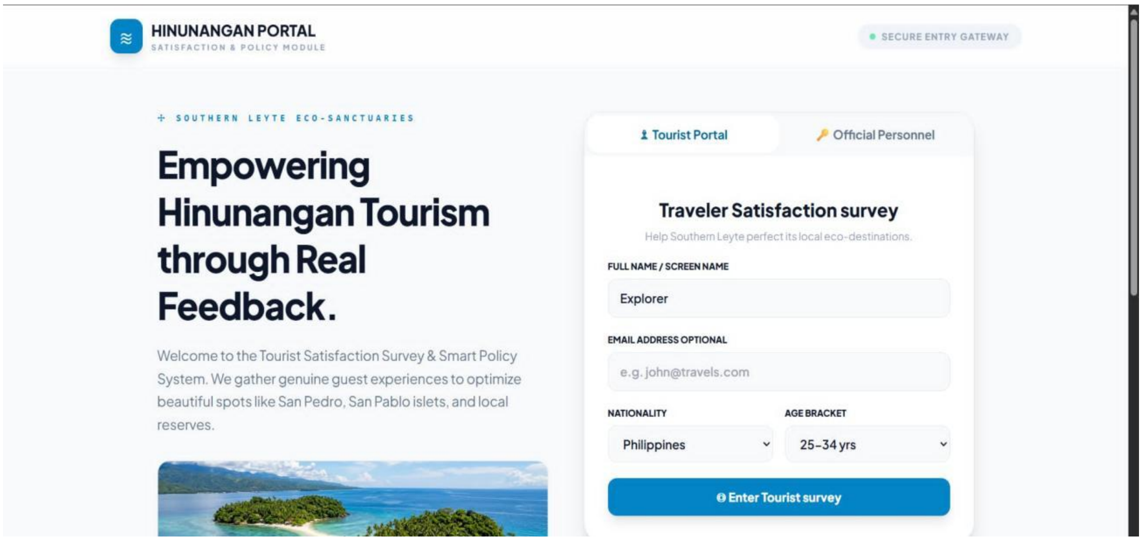
  The database server ensures secure and organized storage of all tourism-related data for analysis and reporting.

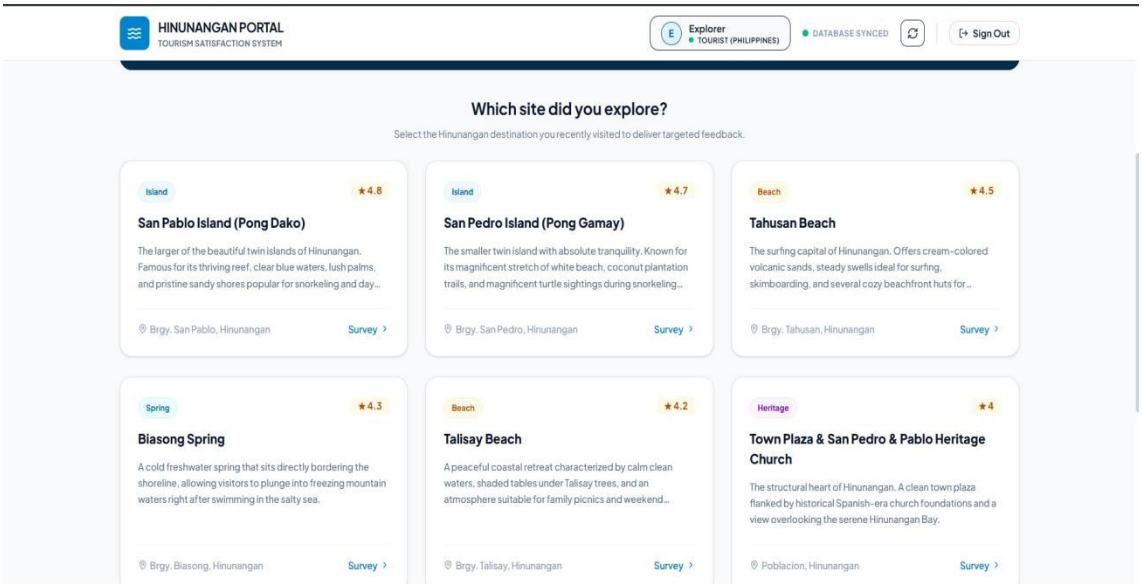
Data Flow

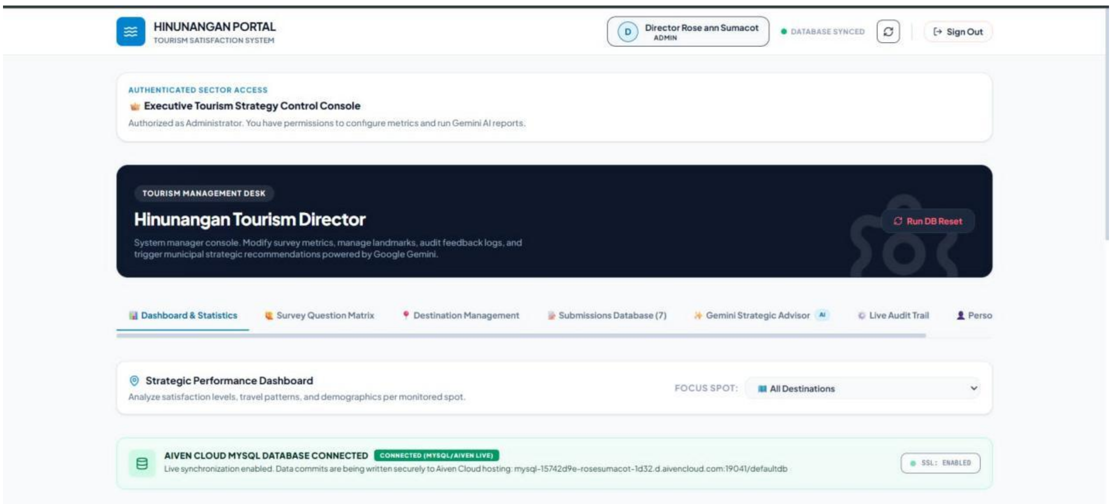
1. Users (Admin, Staff Encoder, Tourist) access the system via a web browser.
2. Requests are sent to the Laravel application server.
3. Laravel processes the request (validation, logic, authentication).
4. Laravel communicates with the MySQL database to store or retrieve data.
5. MySQL returns the requested data to Laravel.
6. Laravel sends the processed response back to the user interface.

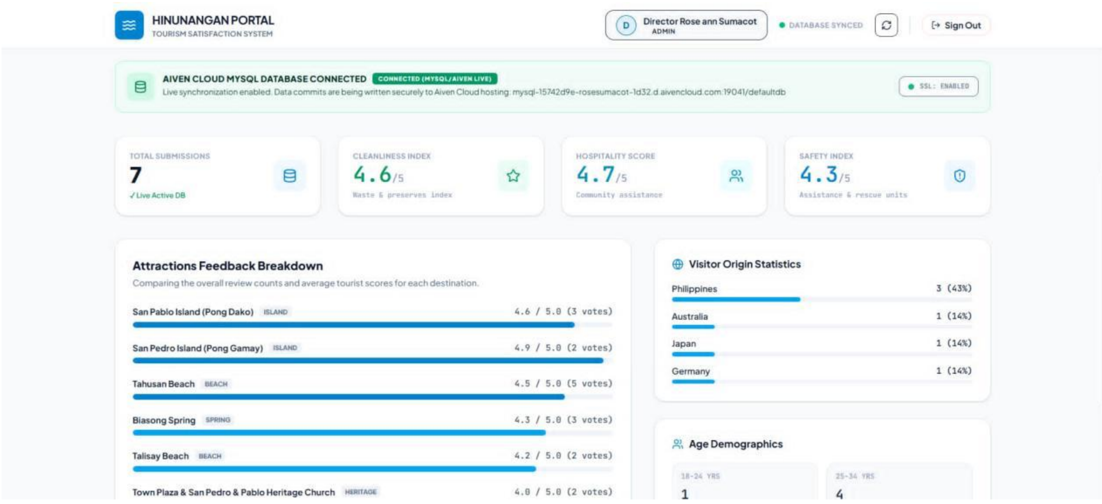

SYSTEM ARCHITECTURE DIAGRAM
HINUNANGAN TOURISM SURVEY SYSTEM
CLIENT SIDE
Web Browser Interface
HINUNANGAN TOURISM SURVEY PORTAL
ADMIN / TOURISM OFFICIAL
• Manage users and roles (e.g., Staff/Analysts)
• System configuration and destination data
• Generate aggregate reports and dashboards
SURVEY STAFF
• Facilitate surveys with tourists
• Data validation and quality control
• Monitor real-time feedback (read-only)
DATA ANALYST
• Deep dive into survey data
• Statistical analysis
• Clean and prepare data for reporting
RESPONDENT / TOURIST
• Access and complete surveys
• Provide feedback on destinations and services
• Opt-in for notifications/reminders
Accessed through Web Browser
HTTP / HTTPS Request
HTTP / HTTPS Response
APPLICATION SERVER
Larevel Framework
Laravel
PHP Framework
Handles System Logic & Requests
Routing
Handles incoming survey requests, staff data, and admin panel routes
Controllers
Process survey submissions, manage user logins, handle report data
Models (Eloquent ORM)
Interact with the database and manage all survey-related data
Services / Survey Logic
Implement core survey flow, data validation, and feedback calculations
Authentication & Authorization
Verify staff/admin users and manage access permissions
Views (Blade Templates)
Render survey forms, dashboards, and reporting interfaces
Database Query
Query Result
DATABASE SERVER
MySQL Database
MySQL
Relational Database
Stores All System Data
• Users (Staff, Admin, Analysts)
• Survey Definitions (Questions, Sections)
• Respondents (Tourists)
• Survey Responses (Data)
• Feedback & Ratings
• Analytics Results
• Tourism Destinations & Attractions
• Notifications / Reminders
• System Logs and Audit Trails
DATA FLOW
1 Users (and tourists) access the system through a web browser.
2 Browser sends HTTP/HTTPS requests (survey submissions, admin commands) to the application server.
3 Laravel application processes the requests, executes survey logic, validates data, and interacts with the database.
4 MySQL database stores and retrieves survey data, responses, and destination info, returning results to Laravel.
5 Laravel sends the response back to the browser and displays dashboards, reports, or the next survey page to the user.

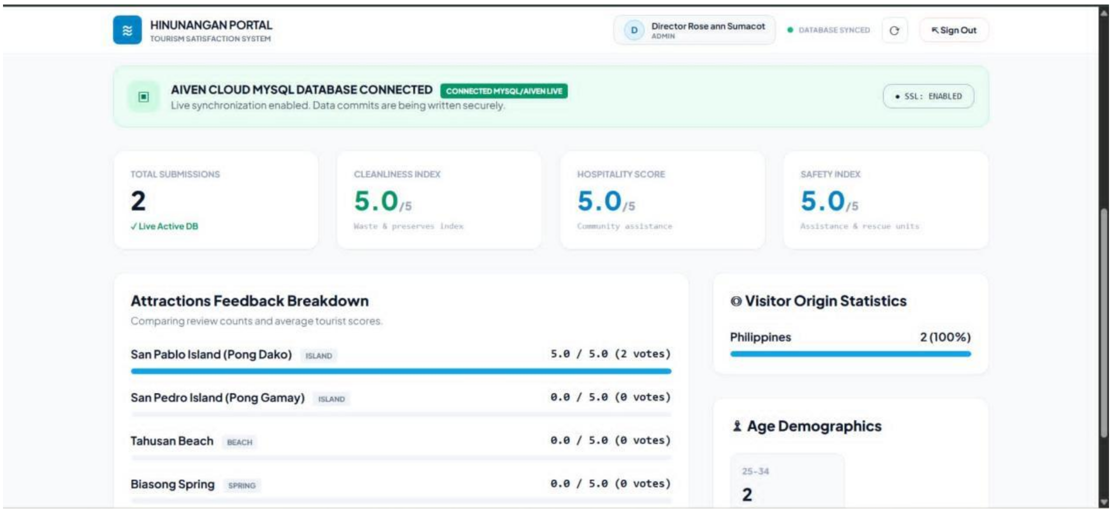

Hinunangan Tourism
Tourist Experience Satisfaction Survey
Start Survey
Official Login
Discover San Pablo, San Pedro, Tahusan & More
Hinunangan Tourist Experience Survey System
A multi-role tourism survey platform for tourists, staff, and administrators. Collect feedback, rate destinations, manage survey questions, and generate tourism analytics reports.
Start Survey
Official Login

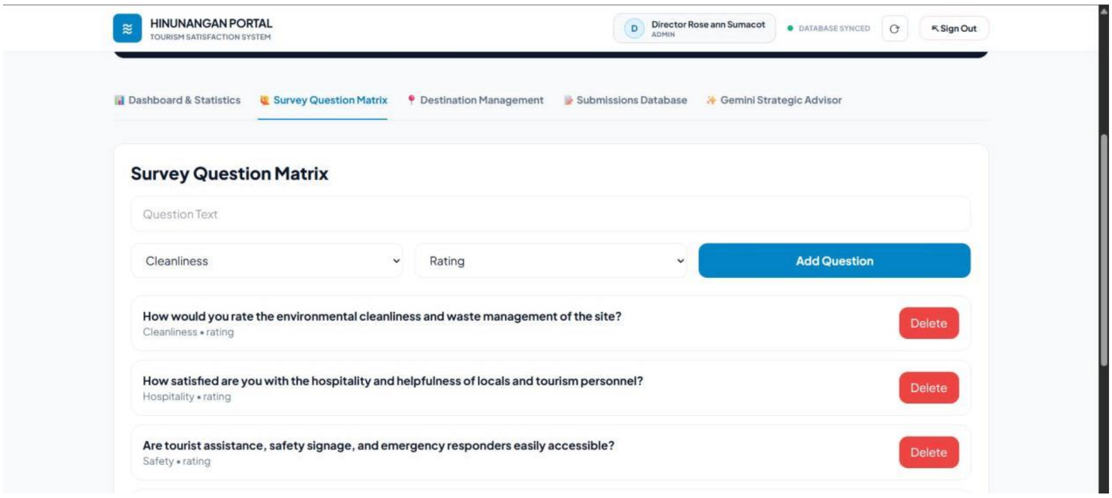

HINUNANGAN PORTAL
SATISFACTION & POLICY MODULE
SECURE ENTRY GATEWAY
SOUTHERN LEYTE ECO-SANCTUARIES
Empowering Hinunangan Tourism through Real Feedback.
Welcome to the Tourist Satisfaction Survey & Smart Policy System. We gather genuine guest experiences to optimize beautiful spots like San Pedro, San Pablo islets, and local reserves.
Tourist Portal
Official Personnel
Traveler Satisfaction survey
Help Southern Leyte perfect its local eco-destinations.
FULL NAME / SCREEN NAME
Explorer
EMAIL ADDRESS OPTIONAL
e.g. john@travels.com
NATIONALITY
Philippines
AGE BRACKET
25-34 yrs
Enter Tourist survey

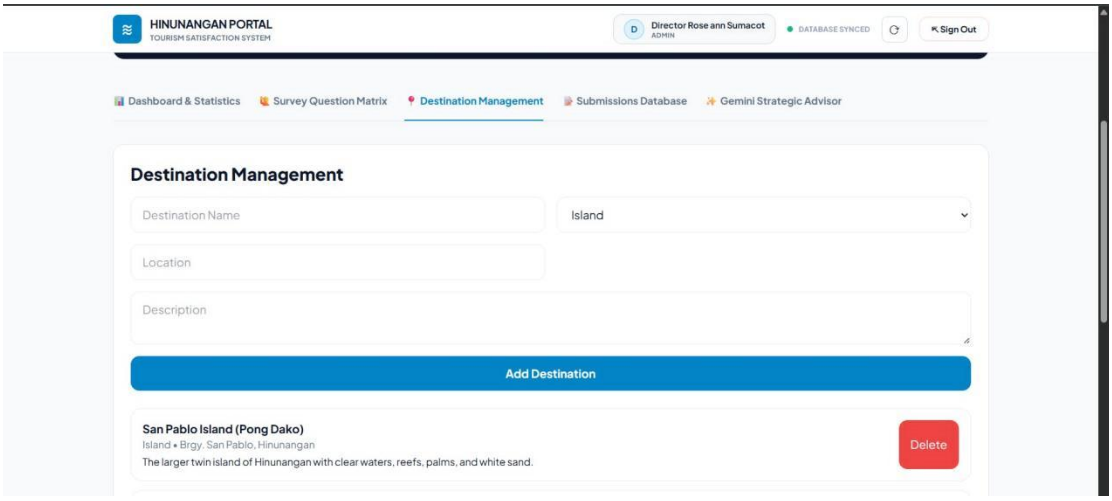

HINUNANGAN PORTAL
TOURISM SATISFACTION SYSTEM
Explorer
TOURIST (PHILIPPINES)
DATABASE SYNCED
Sign Out
Which site did you explore?
Select the Hinunangan destination you recently visited to deliver targeted feedback.
Island
★4.8
San Pablo Island (Pong Dako)
The larger of the beautiful twin islands of Hinunangan. Famous for its thriving reef, clear blue waters, lush palms, and pristine sandy shores popular for snorkeling and day...
Brgy. San Pablo, Hinunangan
Survey
Island
★4.7
San Pedro Island (Pong Gamay)
The smaller twin island with absolute tranquility. Known for its magnificent stretch of white beach, coconut plantation trails, and magnificent turtle sightings during snorkeling...
Brgy. San Pedro, Hinunangan
Survey
Beach
★4.5
Tahusan Beach
The surfing capital of Hinunangan. Offers cream-colored volcanic sands, steady swells ideal for surfing, skimboarding, and several cozy beachfront huts for...
Brgy. Tahusan, Hinunangan
Survey
Spring
★4.3
Biasong Spring
A cold freshwater spring that sits directly bordering the shoreline, allowing visitors to plunge into freezing mountain waters right after swimming in the salty sea.
Brgy. Biasong, Hinunangan
Survey
Beach
★4.2
Talisay Beach
A peaceful coastal retreat characterized by calm clean waters, shaded tables under Talisay trees, and an atmosphere suitable for family picnics and weekend...
Brgy. Talisay, Hinunangan
Survey
Heritage
★4
Town Plaza & San Pedro & Pablo Heritage Church
The structural heart of Hinunangan. A clean town plaza flanked by historical Spanish-era church foundations and a view overlooking the serene Hinunangan Bay.
Poblacion, Hinunangan
Survey

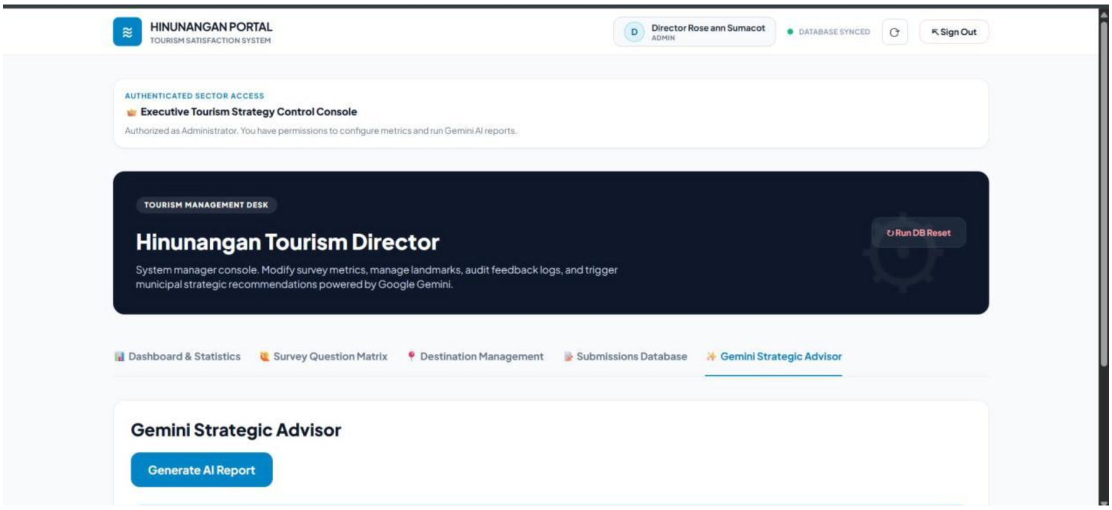

HINUNANGAN PORTAL
TOURISM SATISFACTION SYSTEM
Director Rose ann Sumacot
ADMIN
DATABASE SYNCED
Sign Out
AUTHENTICATED SECTOR ACCESS
Executive Tourism Strategy Control Console
Authorized as Administrator. You have permissions to configure metrics and run Gemini AI reports.
TOURISM MANAGEMENT DESK
Hinunangan Tourism Director
System manager console. Modify survey metrics, manage landmarks, audit feedback logs, and trigger municipal strategic recommendations powered by Google Gemini.
Run DB Reset
Dashboard & Statistics
Survey Question Matrix
Destination Management
Submissions Database (7)
Gemini Strategic Advisor
AI
Live Audit Trail
Perso
Strategic Performance Dashboard
Analyze satisfaction levels, travel patterns, and demographics per monitored spot.
FOCUS SPOT:
All Destinations
AIVEN CLOUD MYSQL DATABASE CONNECTED
CONNECTED (MYSQL/AIVEN LIVE)
Live synchronization enabled. Data commits are being written securely to Aiven Cloud hosting: mysql-15742d9e-rosesumacot-1d32.d.aivencloud.com:19041/defaultdb
SSL: ENABLED

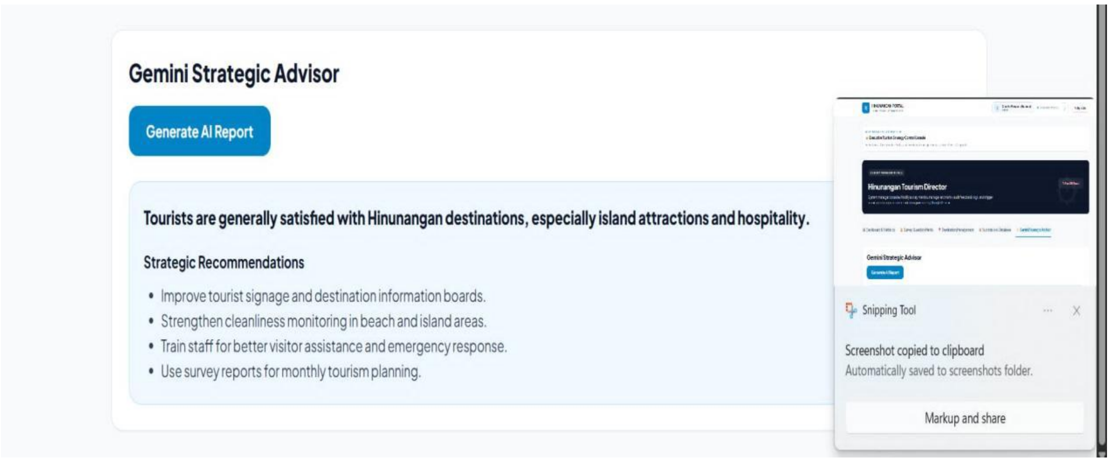

HINUNANGAN PORTAL
TOURISM SATISFACTION SYSTEM
Director Rose ann Sumacot
ADMIN
DATABASE SYNCED
Sign Out
AIVEN CLOUD MYSQL DATABASE CONNECTED
CONNECTED (MYSQL/AIVEN LIVE)
Live synchronization enabled. Data commits are being written securely to Aiven Cloud hosting: mysql-15742d9e-rosesumacot-1d32.d.aivencloud.com:19041/defaultdb
SSL: ENABLED
TOTAL SUBMISSIONS
7
✓ Live Active DB
CLEANLINESS INDEX
4.6/5
Waste & preserves index
HOSPITALITY SCORE
4.7/5
Community assistance
SAFETY INDEX
4.3/5
Assistance & rescue units
Attractions Feedback Breakdown
Comparing the overall review counts and average tourist scores for each destination.
San Pablo Island (Pong Dako) ISLAND 4.6 / 5.0 (3 votes)
San Pedro Island (Pong Gamay) ISLAND 4.9 / 5.0 (2 votes)
Tahusan Beach BEACH 4.5 / 5.0 (5 votes)
Biasong Spring SPRING 4.3 / 5.0 (3 votes)
Talisay Beach BEACH 4.2 / 5.0 (2 votes)
Town Plaza & San Pedro & Pablo Heritage Church HERITAGE 4.0 / 5.0 (2 votes)
Visitor Origin Statistics
Philippines 3 (43%)
Australia 1 (14%)
Japan 1 (14%)
Germany 1 (14%)
Age Demographics
18-24 YRS
1
25-34 YRS
4


HINUNANGAN PORTAL
TOURISM SATISFACTION SYSTEM
Director Rose ann Sumacot
ADMIN
DATABASE SYNCED
Sign Out
AIVEN CLOUD MYSQL DATABASE CONNECTED
CONNECTED MYSQL/AIVEN LIVE
Live synchronization enabled. Data commits are being written securely.
SSL: ENABLED
TOTAL SUBMISSIONS
2
✓ Live Active DB
CLEANLINESS INDEX
5.0/5
Waste & preserves index
HOSPITALITY SCORE
5.0/5
Community assistance
SAFETY INDEX
5.0/5
Assistance & rescue units
Attractions Feedback Breakdown
Comparing review counts and average tourist scores.
San Pablo Island (Pong Dako) ISLAND 5.0 / 5.0 (2 votes)
San Pedro Island (Pong Gamay) ISLAND 0.0 / 5.0 (0 votes)
Tahusan Beach BEACH 0.0 / 5.0 (0 votes)
Biasong Spring SPRING 0.0 / 5.0 (0 votes)
Visitor Origin Statistics
Philippines 2 (100%)
Age Demographics
25-34
2


HINUNANGAN PORTAL
TOURISM SATISFACTION SYSTEM
Director Rose ann Sumacot
ADMIN
DATABASE SYNCED
Sign Out
Dashboard & Statistics
Survey Question Matrix
Destination Management
Submissions Database
Gemini Strategic Advisor
Survey Question Matrix
Question Text
Cleanliness
Rating
Add Question
How would you rate the environmental cleanliness and waste management of the site?
Cleanliness • rating
Delete
How satisfied are you with the hospitality and helpfulness of locals and tourism personnel?
Hospitality • rating
Delete
Are tourist assistance, safety signage, and emergency responders easily accessible?
Safety • rating
Delete


HINUNANGAN PORTAL
TOURISM SATISFACTION SYSTEM
Director Rose ann Sumacot
ADMIN
DATABASE SYNCED
Sign Out
Dashboard & Statistics
Survey Question Matrix
Destination Management
Submissions Database
Gemini Strategic Advisor
Destination Management
Destination Name
Island
Location
Description
Add Destination
San Pablo Island (Pong Dako)
Island • Brgy. San Pablo, Hinunangan
The larger twin island of Hinunangan with clear waters, reefs, palms, and white sand.
Delete


HINUNANGAN PORTAL
TOURISM SATISFACTION SYSTEM
Director Rose ann Sumacot
ADMIN
DATABASE SYNCED
Sign Out
AUTHENTICATED SECTOR ACCESS
Executive Tourism Strategy Control Console
Authorized as Administrator. You have permissions to configure metrics and run Gemini AI reports.
TOURISM MANAGEMENT DESK
Hinunangan Tourism Director
System manager console. Modify survey metrics, manage landmarks, audit feedback logs, and trigger
municipal strategic recommendations powered by Google Gemini.
Run DB Reset
Dashboard & Statistics
Survey Question Matrix
Destination Management
Submissions Database
Gemini Strategic Advisor
Gemini Strategic Advisor
Generate AI Report


Gemini Strategic Advisor
Generate AI Report
Tourists are generally satisfied with Hinunangan destinations, especially island attractions and hospitality.
Strategic Recommendations
• Improve tourist signage and destination information boards.
• Strengthen cleanliness monitoring in beach and island areas.
• Train staff for better visitor assistance and emergency response.
• Use survey reports for monthly tourism planning.
Hinunangan Tourism Director
Gemini Strategic Advisor
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Markup and share

Database Structure

1. Users Table

Stores system user accounts

Field	Type	Description
id	INT	Primary key
name	VARCHAR	User name
email	VARCHAR	Email address
password	VARCHAR	Encrypted password
role	VARCHAR	Admin/User
created_at	TIMESTAMP	Date created
updated_at	TIMESTAMP	Date updated

2. Surveys Table

Stores survey information

Field	Type	Description
survey_id	INT	Primary key
title	VARCHAR	Survey title
description	TEXT	Survey description
created_at	TIMESTAMP	Date created

3. Questions Table

Stores survey questions

Field	Type	Description
question_id	INT	Primary key

Field	Type	Description
survey_id	INT	Linked survey
question_text	TEXT	Question content

4. Responses Table

Stores survey answers

Field	Type	Description
response_id	INT	Primary key
survey_id	INT	Linked survey
answer	TEXT	User response
created_at	TIMESTAMP	Date submitted

Relationships Between Tables

- One **User account** can be assigned as **Admin, Staff Encoder, or Tourist/User**.
- One **Survey** can contain **multiple Questions**.
- One **Tourist/User** can submit **multiple Survey Responses**.
- One **Survey** can receive **many Responses from different users**.
- Each **Response** is linked to a specific **Survey and Question** through IDs (survey_id, question_id).
- The **Admin and Staff Encoder** manage surveys, questions, and monitor all submitted responses.
- All relationships ensure organized storage and easy retrieval of tourism feedback data.

Implementation and Testing

Implementation

The system was developed using the Laravel framework with PHP and MySQL. The development process included system planning, database design, coding, and interface development.

The frontend was created using HTML, CSS, JavaScript, and Bootstrap to ensure a responsive design. The backend handles survey submission, data processing, and reporting.

The system allows tourists to answer surveys without needing accounts, while administrators manage and analyze results through a secure dashboard.

Testing

Functional Testing

- Survey submission tested successfully
- Admin login and dashboard verified
- Data storage checked and validated

Database Testing

- Ensured correct saving of responses
- Verified relationships between tables
- Tested data retrieval accuracy

UI Testing

- Checked responsiveness on different devices
- Verified form validation
- Ensured proper navigation flow

Problems Encountered

- Errors in database configuration
- Issues in form validation
- Minor bugs in data display

Solutions Applied

- Fixed Laravel environment configuration
- Improved validation rules
- Debugged and tested system repeatedly

Conclusion

The Hinunangan Tourism Survey System was successfully developed to improve the collection and management of tourism feedback in Hinunangan. The system provides a more efficient, organized, and reliable way of gathering tourist opinions compared to traditional paper-based methods.

It helps tourism officers easily access survey results and supports better decision-making for improving local tourism services. The system also reduces manual work and improves data accuracy.

Summary

The system integrates tourism survey collection into a centralized web platform. It improves efficiency, accuracy, and accessibility of tourism feedback, making it easier for officials to analyze data and improve services.

What Was Achieved in the Project

The project successfully created a digital tourism survey system that allows tourists to submit feedback online and enables administrators to manage and analyze responses efficiently.

Benefits of the System

- Faster data collection
- Reduced paperwork
- Improved data accuracy
- Easier analysis of tourism feedback
- Better tourism planning and decision-making

Learning Gained During Development

The development of this system improved skills in Laravel, PHP, MySQL, HTML, CSS, and JavaScript. It also enhanced problem-solving, system design, and database management skills.

Recommendations

Future Improvements

- Develop a mobile application version
- Add offline survey capability
- Improve UI/UX design

Additional Features

- Email/SMS notifications
- Advanced analytics dashboard

- Multi-language support for tourists
- QR code-based survey access

System Enhancements

- AI-based sentiment analysis of responses
- Integration with tourism booking systems
- Cloud-based deployment for scalability
- Real-time reporting dashboa